

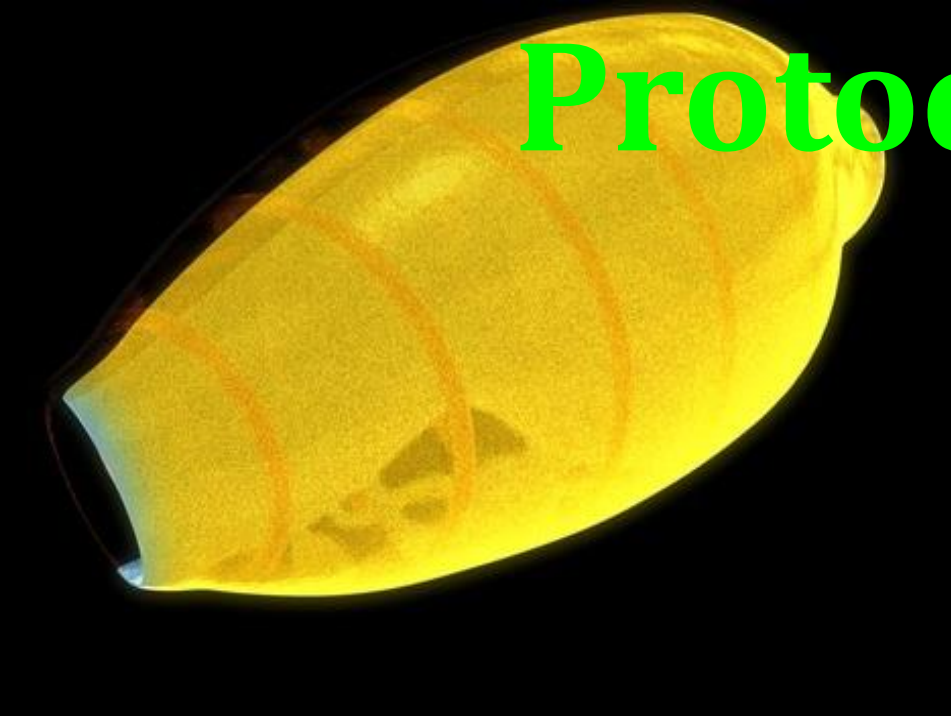


ZOO 1302 Vertebrate Diversity

Dr. (Mrs.) T.V. Sundarabharathy
Department of Biology
Faculty of Applied Sciences
Rajarata University of Sri Lanka



Lesson 3



Protochordates

Learning Outcomes

At the end of the lesson you would be able to describe the

- general characteristics of sub phyla of protochordates.
- characteristics features of *Ciona*.
- characteristics features of *Amphioxus*.

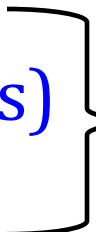
Phylum - Chordata

Chordates are divided into three basic groups:

1. Sub phylum - Urochordata (tunicates)

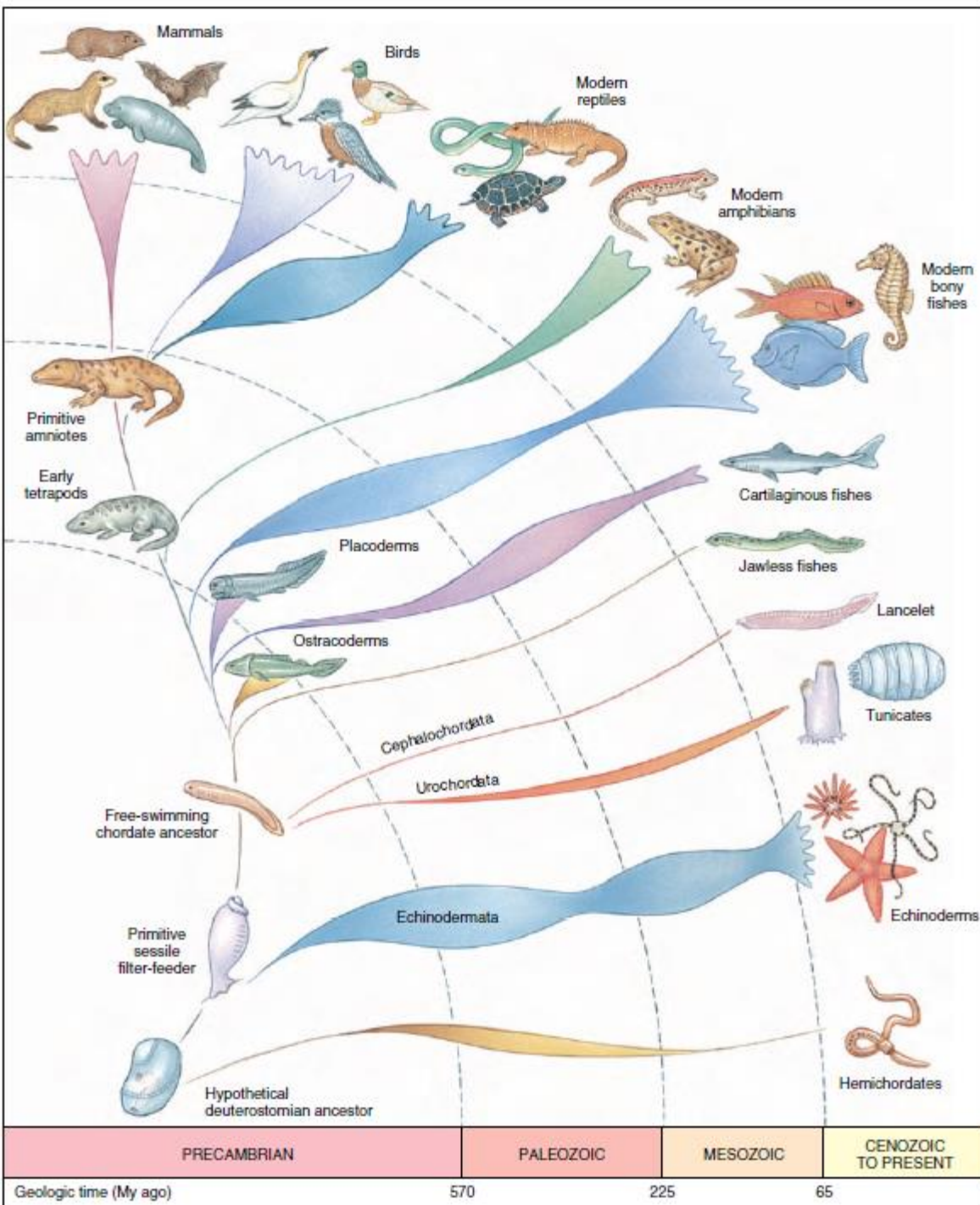
2. Sub phylum - Cephalochordata (lancelets)

3. Sub phylum – Vertebrata (Craniates)

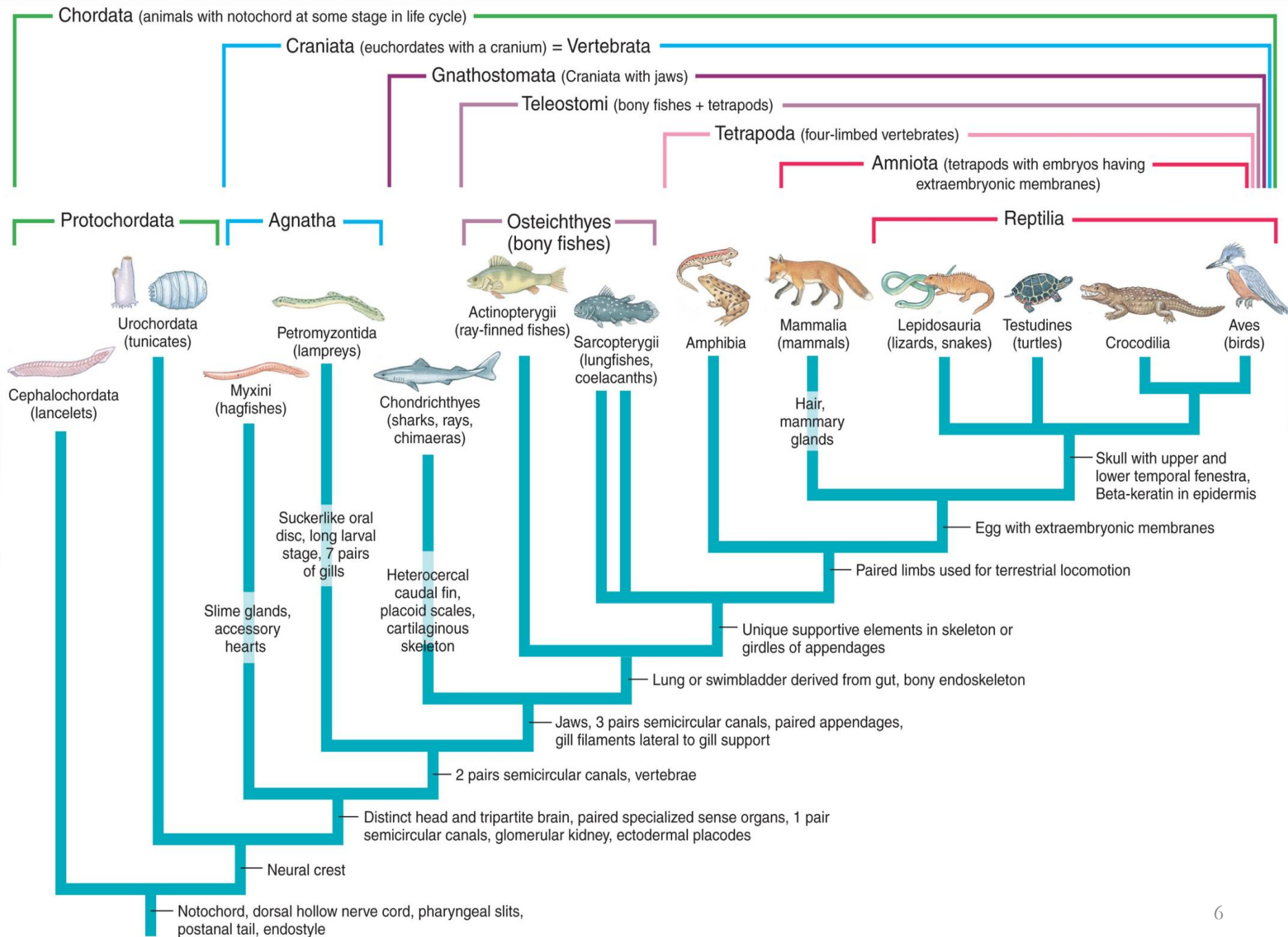


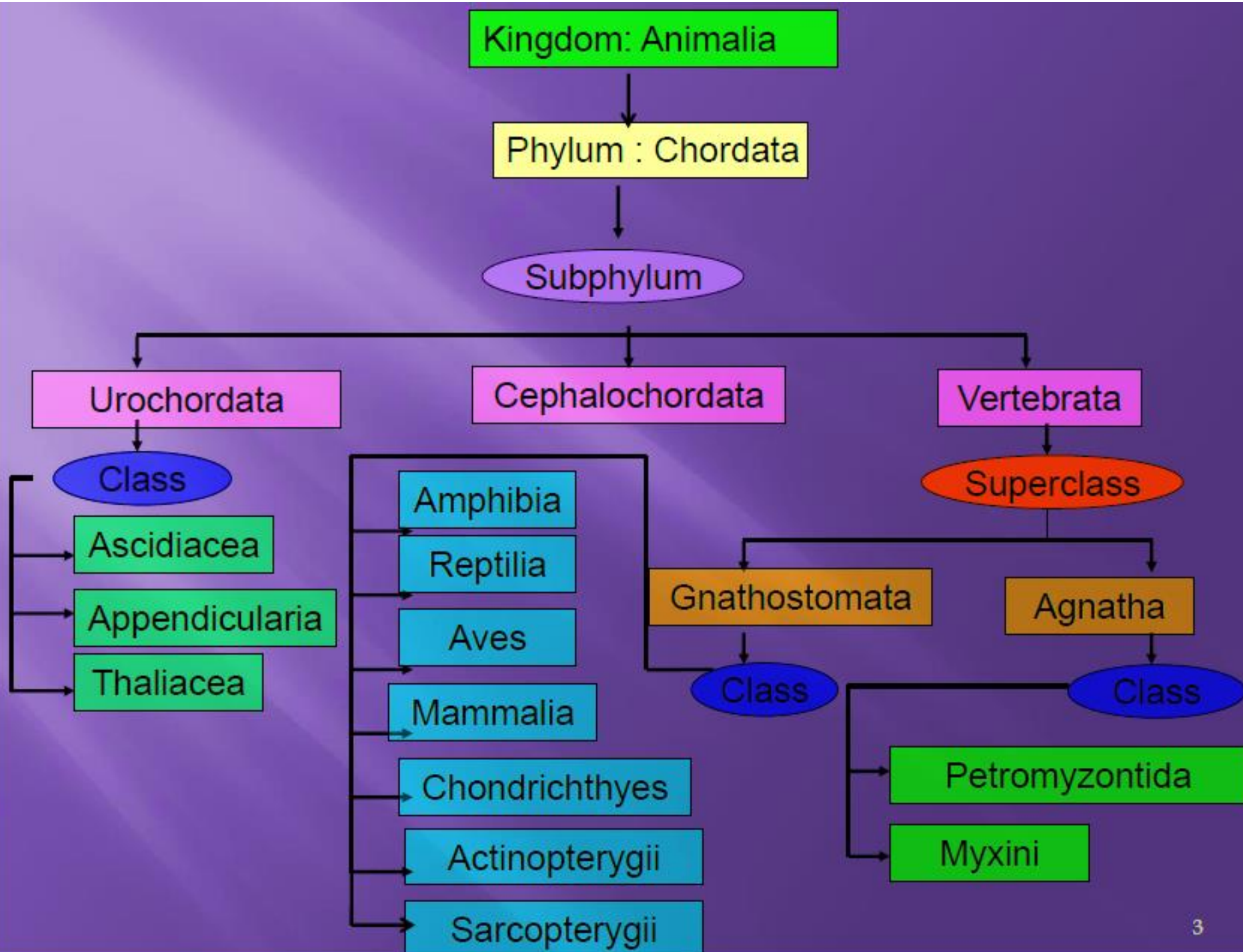
Protochordates

- Of these, the **vertebrates** - *lampreys, mammals, birds, amphibians, reptiles and fishes* — are the most familiar and are the group to which humans belong.



Phylogenetic tree of the chordates, suggesting probable origin and relationships





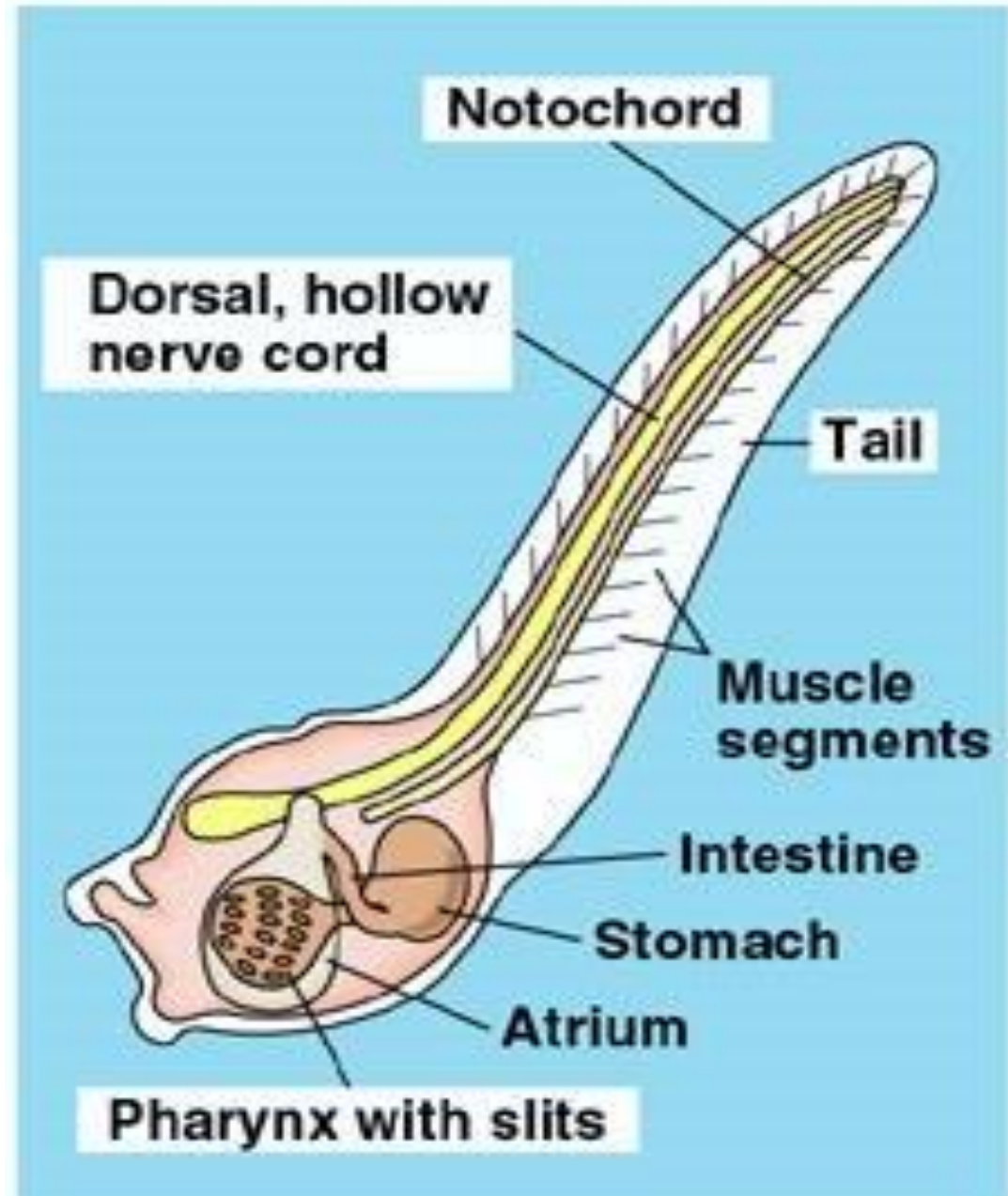
Protochordates

- Prochordates are very primitive chordates which shows **five** chordate hall marks at least in their larval stage.
- They are divided into 2 sub phyla under Phylum – Chordata.
 1. Sub phylum - Urochordata (tunicates)
 2. Sub phylum - Cephalochordata (lancelets)

Sub Phylum : Urochordata (Tunicates)

- Commonly called tunicates
- Found in seas from near shoreline to great depths
- Most are sessile as adults , some are free living
- Some species are solitary, others are colonial
- **Tunicate** - animal is usually surrounded by tough, nonliving tunic which contains cellulose
- Tunicates are filter feeders

- All chordate characters present
- 3 classes
 1. class- Ascidiacea
 2. class - Thaliacea
 3. class – Larvacea (Appendicularia)



Subphylum - Urochordata

Class

Ascidiacea

Appendiculari

Thaliacea



Ciona sp.



Oikopleura



Doliolum

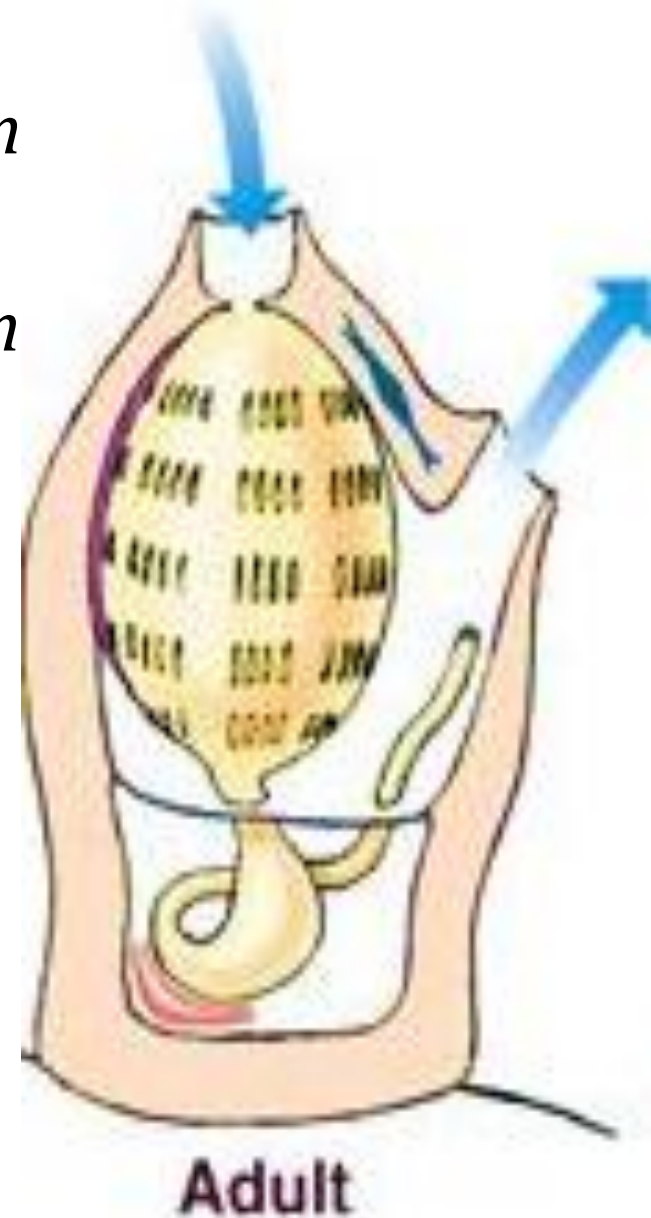


Class – Ascidiacea

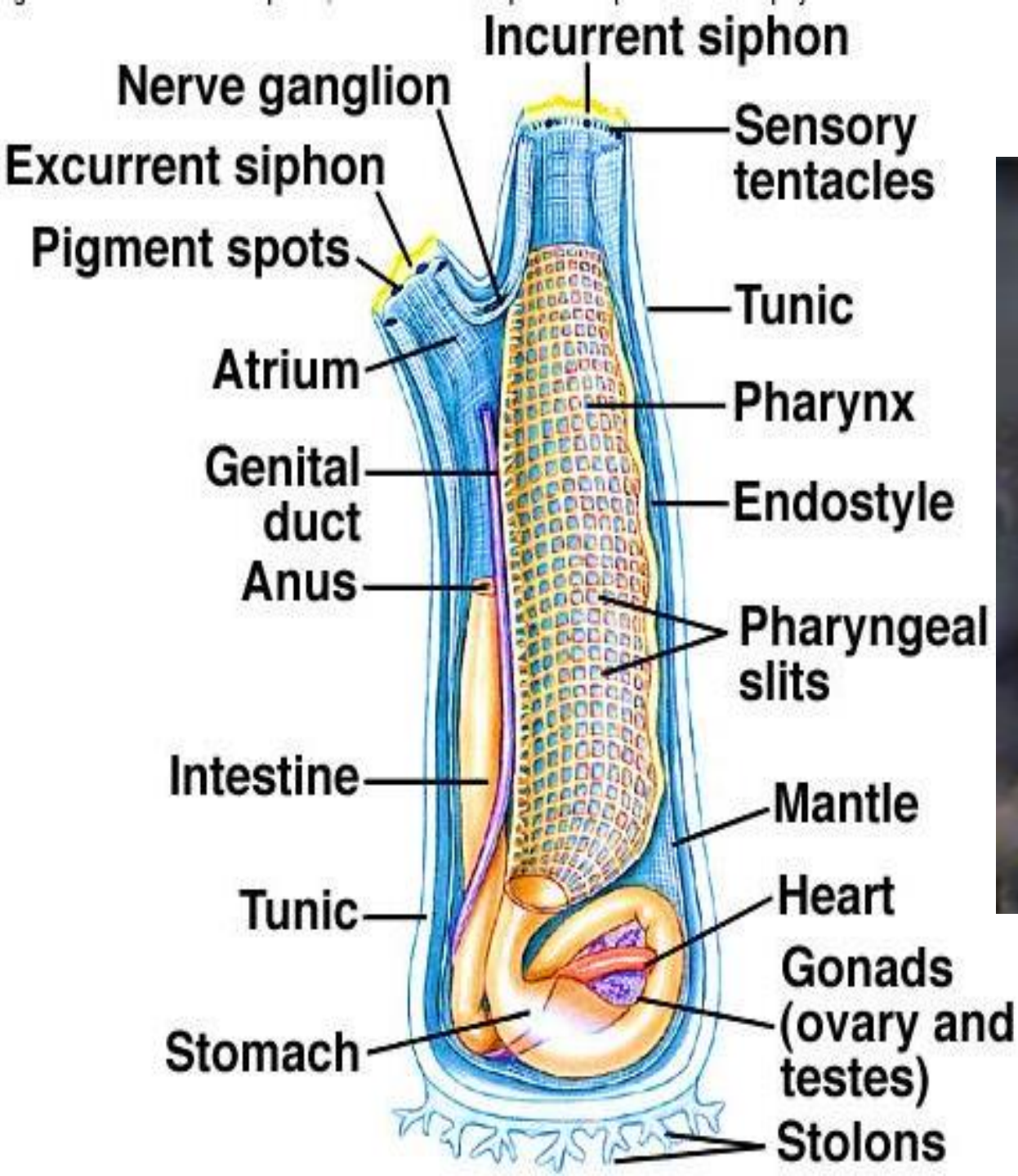
Class – Ascidiacea Eg. *Ciona*

- *Askiolion* – little bag + *ascea* – suffix.
- Often called as Sea squirts - when irritated, forcefully discharge a jet of water from the excurrent siphon.
- Few are sessile – attached to rock or other hard substrates.
- Solitary, colonial or compound and cylindrical.
- Outer covering - tunic
- Lining the tunic is an inner membrane – mantle
- Each of solitary and colonial forms has its own test, but among the compound form many individuals may share the same test.

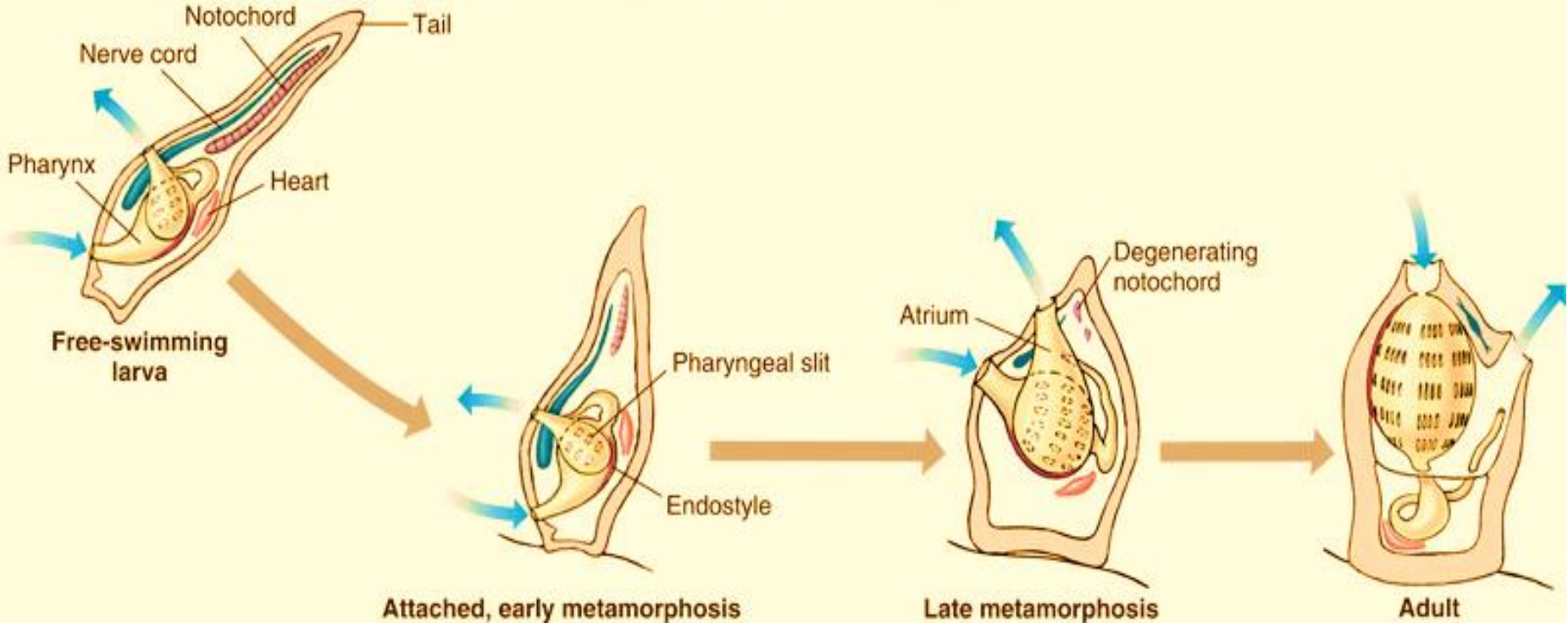
- Two projections on the outside
 - ✓ an *incurrent siphon* or *oral siphon*
– anterior end
 - ✓ an *excurrent siphon* or *atrial siphon*
– dorsal side
- When sea squirt expands, water enters the incurrent siphon.
- Then the water passes into ciliated pharynx- is a pharyngeal basket which is minutely perforated with slits.
- Water passes through the slits into an atrial cavity and out through the excurrent siphon.



- In some compound ascidians - each member has its own incurrent siphon, but the excurrent opening is common to the groups.
- Feeding - depends on formation of a mucous net that is secreted by a glandular groove - endostyle
- Endostyle – located along the midventral side of the pharynx
- Circulatory system – ventral heart and 2 large vessels on either side of the heart
- Nervous system – restricted to a ganglion & plexus of nerves – lie on the dorsal side of the pharynx
- Subneural gland – located beneath the nerve ganglion

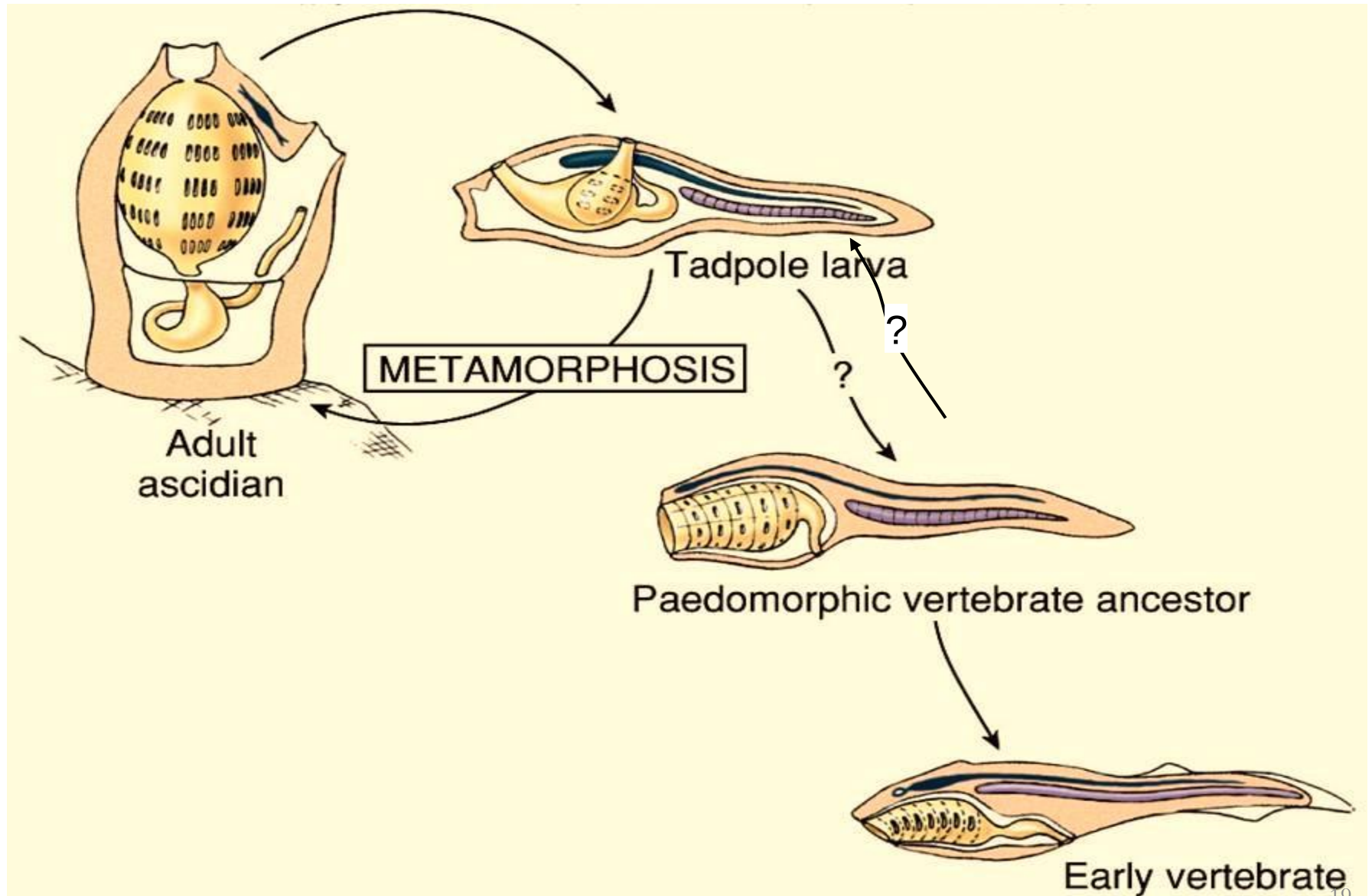


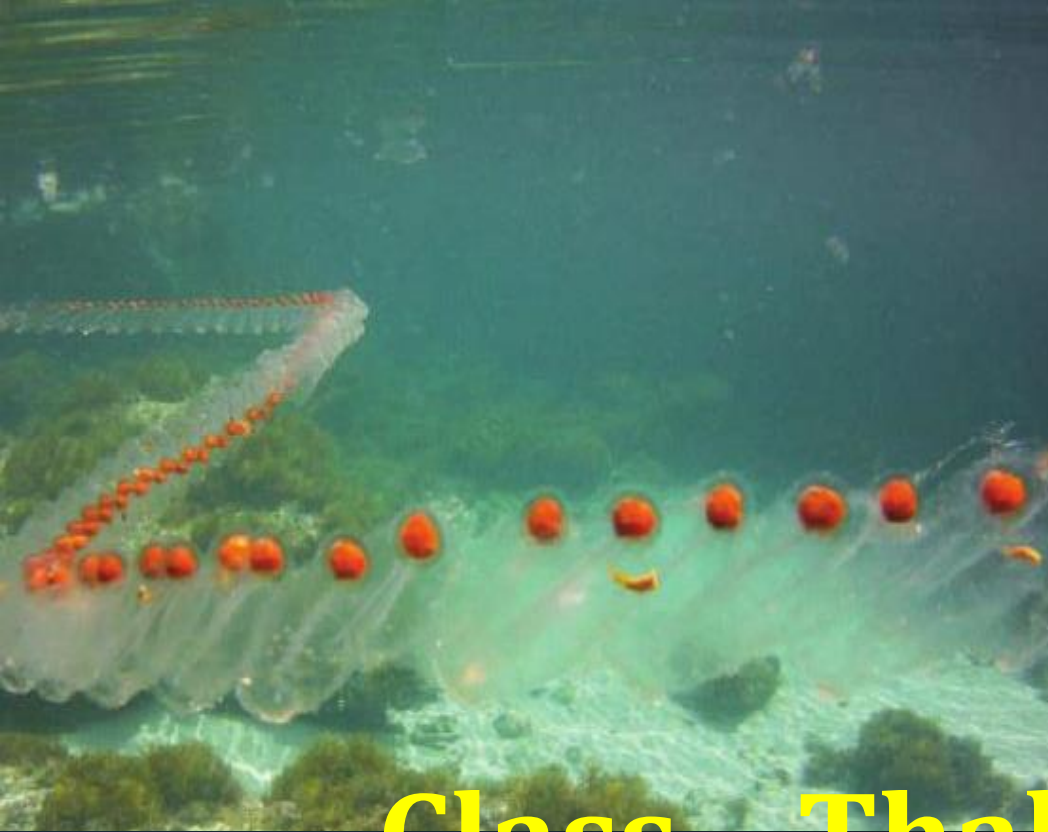
- Reproduction – hermaphrodites , possess single ovary and single testis
- Germ cells are carried by ducts into the atrial cavity , and send out to the surrounding water
- Fertilization – external
- Larva – a tadpole larva , does not feed
- Larva swims for some hours or days and by adhesive papillae the fasten to a solid object
- Then undergo metamorphosis – become sessile adult



- Tadpole of sea squirts – have all 5 chordate characters
- During metamorphosis - tail, notochord, muscle segments, and nerve cord disappear
- Adults have only – endostyle, pharyngeal slits

The Ancestral Chordate



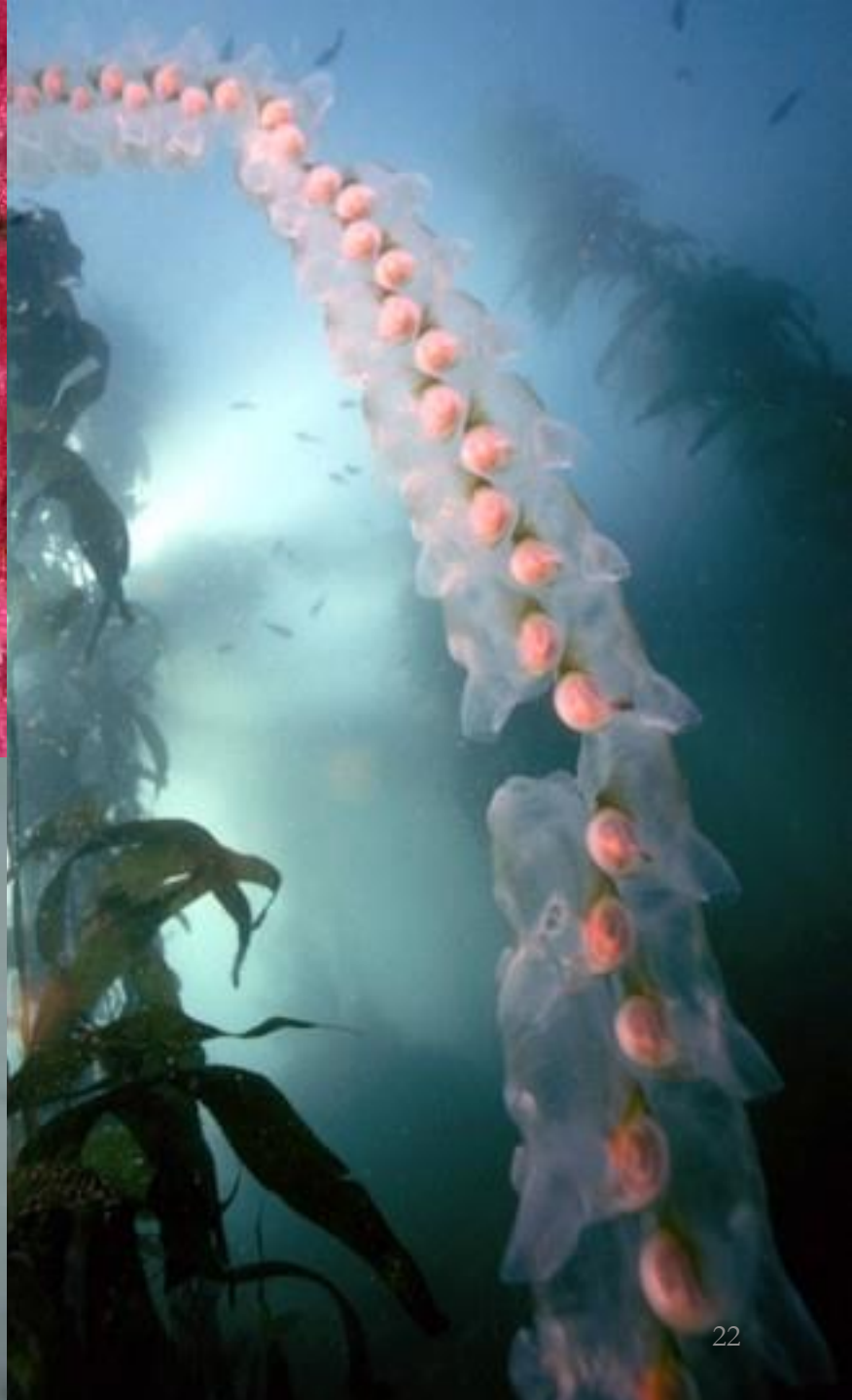


Class - Thaliacea

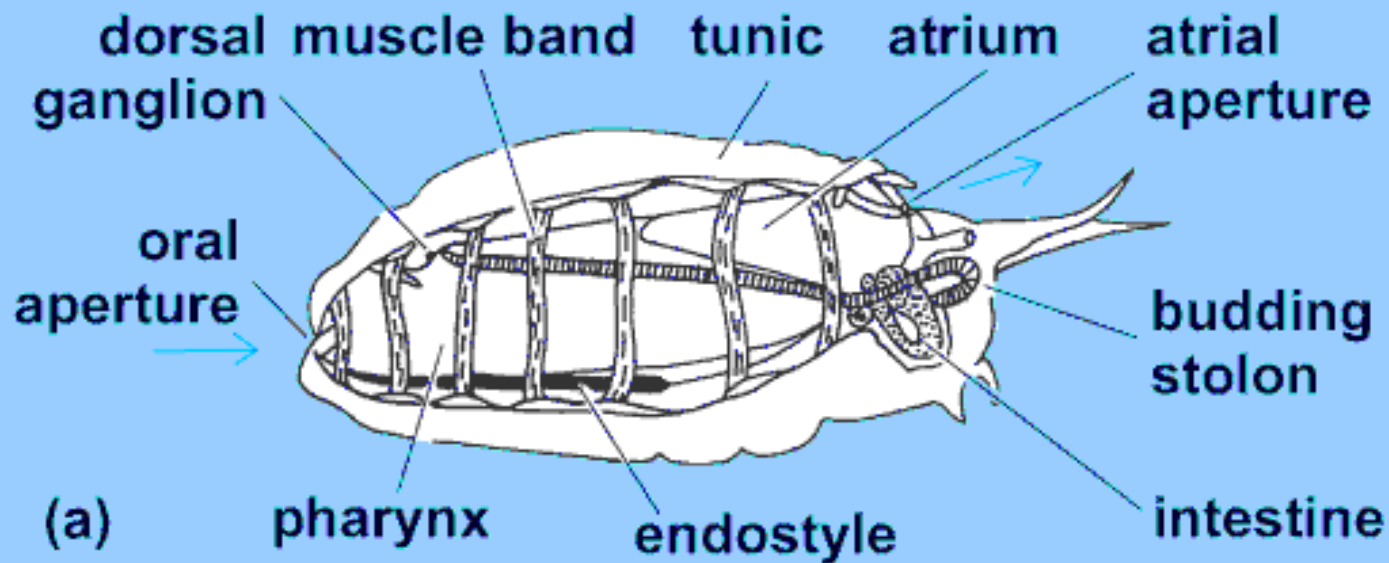


Class – Thaliacea

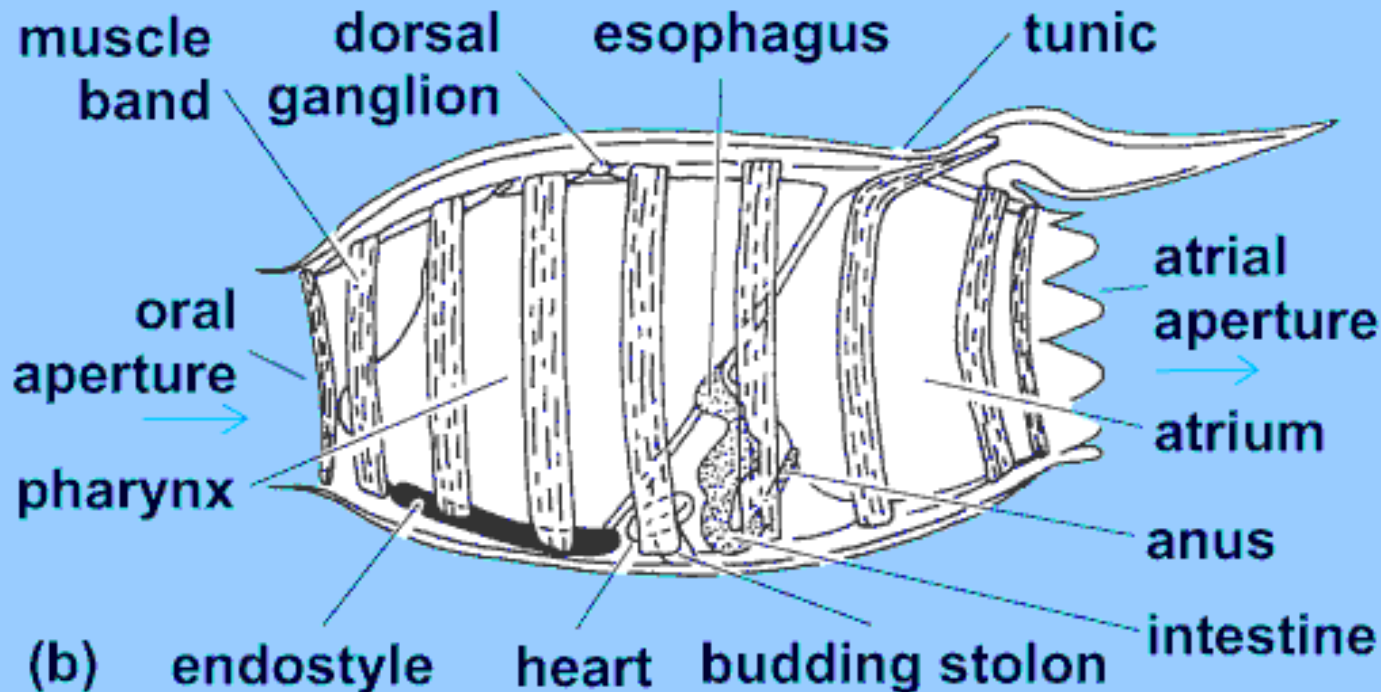
- Eg. *Doliolum* , *Salpa*
- Known as thaliaceans
- Habitat - pelagic forms
- Barrel or lemon shaped with transparent gelatinous bodies
- Nearly invisible in sunlit surface water
- Occur single or in colonial chains
- Cylindrical body, surrounded by bands of circular muscles



- Incurrent(buccal) and excurrent(atrial) siphons at opposite ends
- Water pumped through the body through muscular contraction
- Water pump used in.....
 - locomotion - jet propulsion
 - for respiration
 - as a source of particulate food – filtered on mucous surface



Salpa



Doliolum



Salpa



Doliolum



- Many have luminous organ – give a brilliant light at night
- Body is mostly hollow
- Viscera form a compact mass on the ventral side
- Reproduction – reproduce alternatively sexual and asexual generations

A photograph of a large group of Appendicularia (Larvacea) swimming in clear blue water. The organisms are transparent, elongated, and have a distinctive red, bulbous head at one end. They are arranged in a loose, diagonal line, with some individuals appearing to be in different stages of movement or feeding. The background is a deep blue, suggesting an underwater environment.

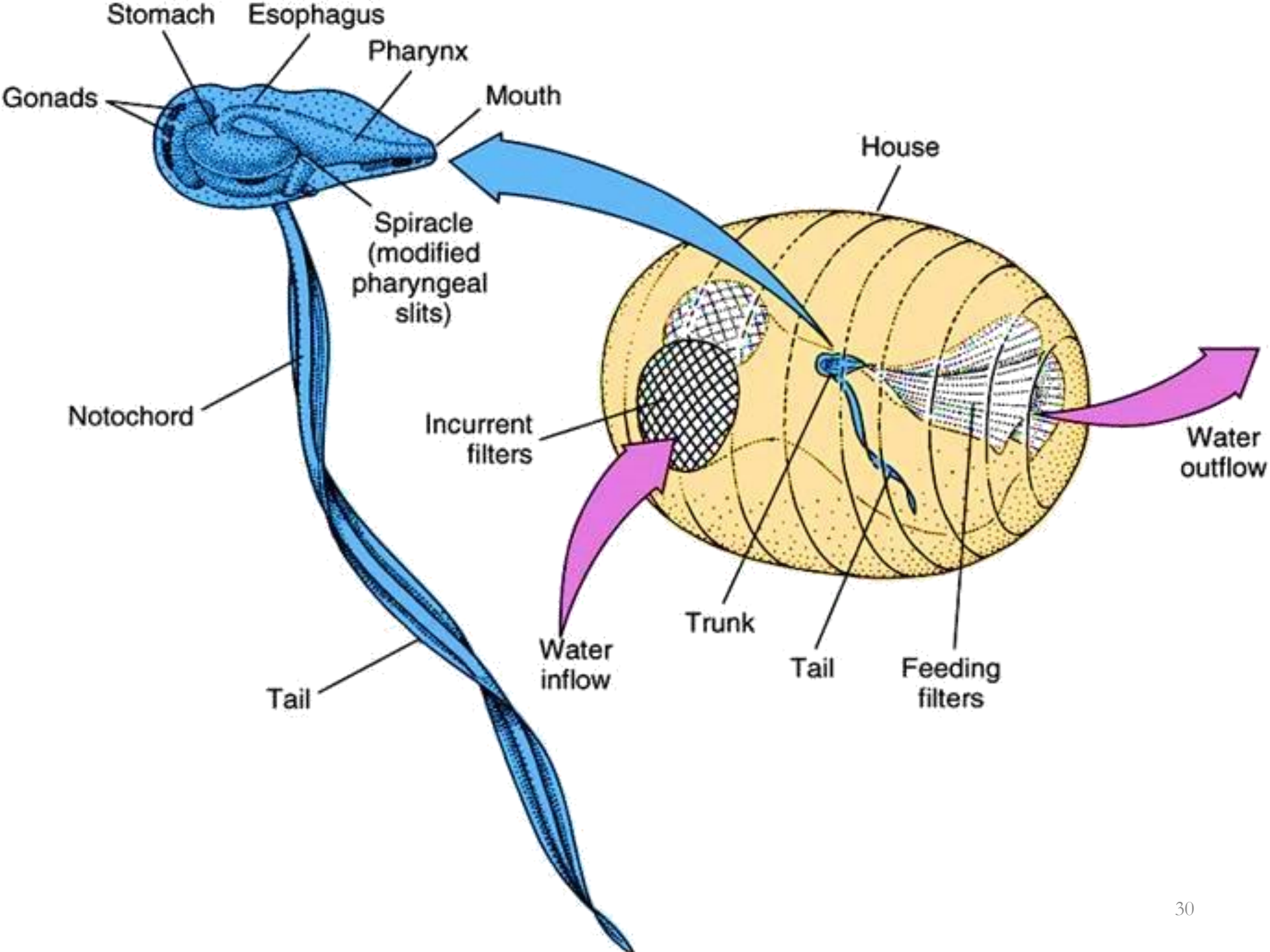
Class – Appendicularia (Larvacea)

Class – Appendicularia

- Larva like pelagic creatures – a bent tadpole shaped
- Resemblance to larva stage of other tunicates
- Each builds ...
 - ✓ a delicate house,
 - ✓ a transparent hollow sphere
 - ✓ of mucus interlaced with filters
- eg. *Oikopleura*

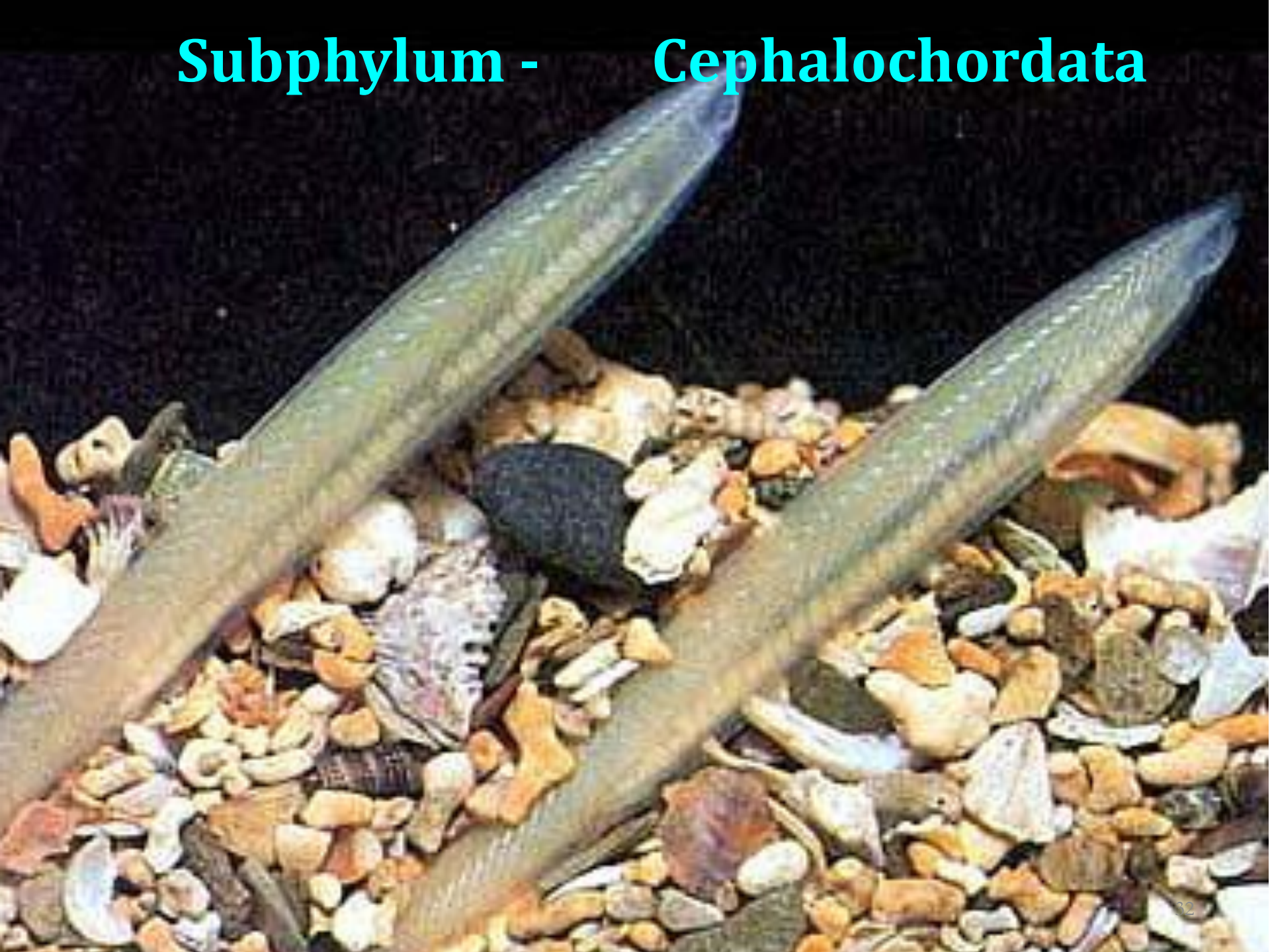


- Tiny phytoplakton and bacteria trapped on a feeding filter inside the house – drawn into animals mouth through a strawlike tube.
- The filters become clogged with waste which happens about every 4 hours.
- Then the animal leave abandons its house and builds a new house – this process takes only a few minutes.



- Shows paedomorphosis – the development of ancestral larval or juvenile features into descendant adult
- They are sexually mature animals that have retained the larval body form.

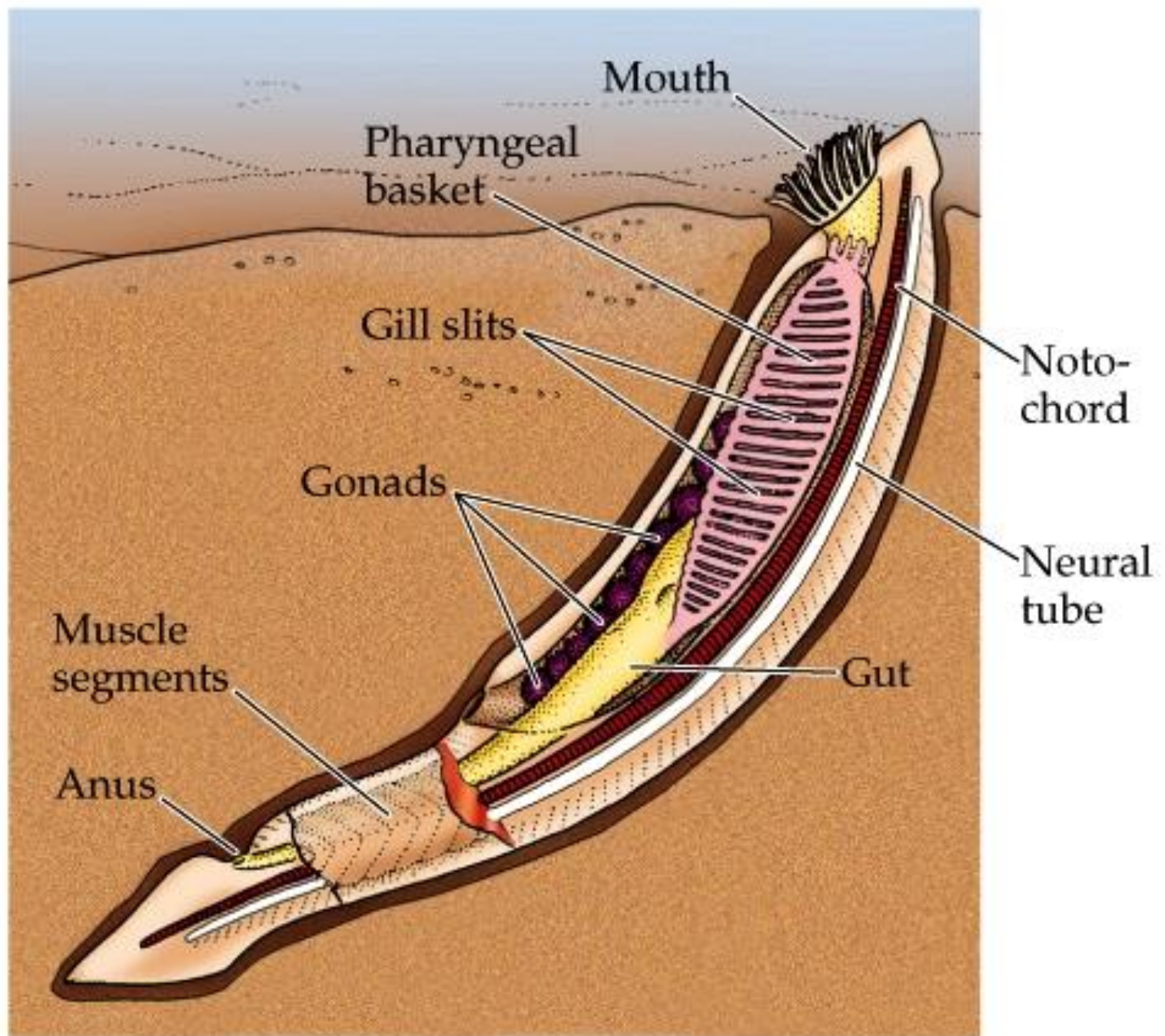
Subphylum - Cephalochordata



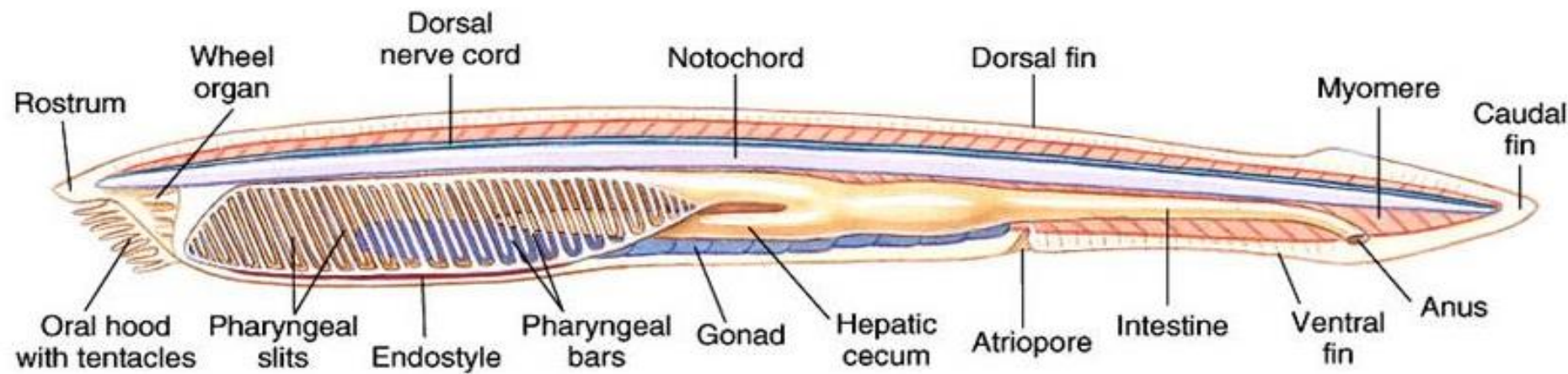
- Filter feeding is accomplished using pharyngeal slits and a mucous net secreted by the endostyle.
- The dorsal, hollow nerve cord lies just above the notochord.
- The circulatory system is closed, but there is no heart.
- Blood functions in nutrient transport, little role in oxygen transport.

Subphylum - Cephalochordata (lancelates)

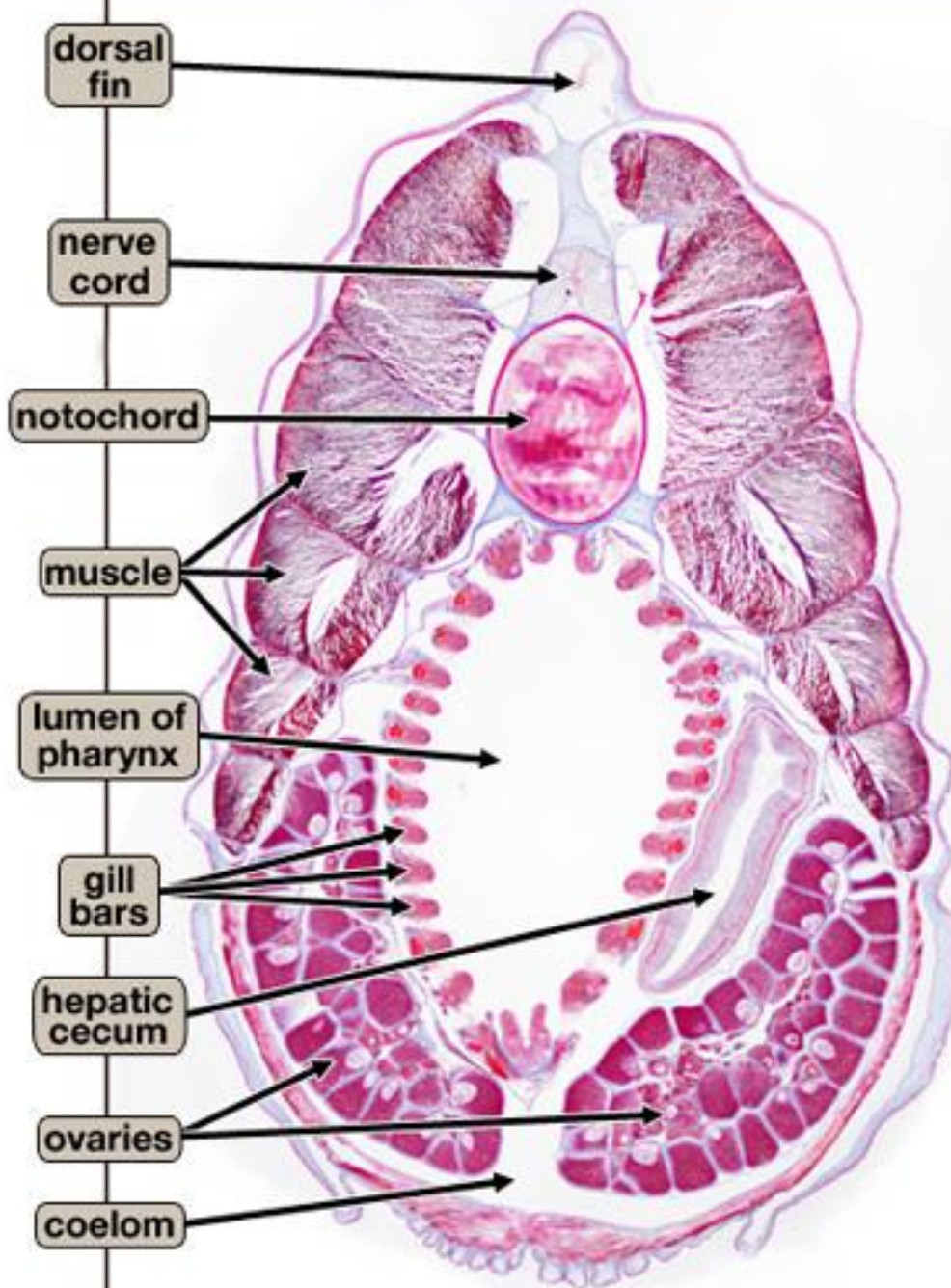
- All five chordate characters are present in a simple form.
- The cephalochordates are the lancelets – both ends are sharp.
- Small (3-7 cm long) laterally compressed fishlike animals.
- Inhabit sandy bottoms of coastal waters.
- They lack a distinct head and have no cranium.



- They are commonly referred to as *Amphioxus* as this was the original genus name.
- There are 29 species, five of which occur in North American coastal waters.
- Have a transparent body.
- Closed circulatory system.
- Hollow nerve cord lies above the notochord.



B



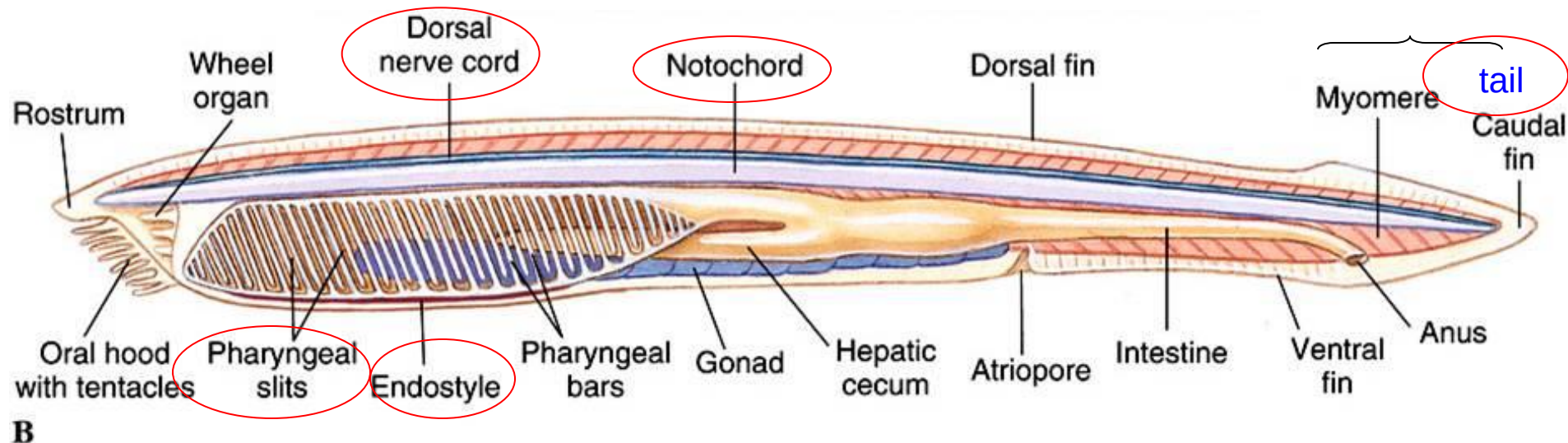
Branchiostoma, c.s., gonad

- Pairs of spinal nerve roots emerge at each trunk myomeric segment.
- Anterior end do not develop into a brain but it is the homologous part of the vertebrate brain.
- Amphioxus is a filter feeder.
- Water enters the mouth and then is moved by beating cilia through the pharyngeal slits, where food is trapped in mucus.
- Cilia then move the food to the gut.

- The smaller food particles are separated from mucus and passed to hepatic caecum and digested.
- Filtered water passes to atrium and then leaves the body through atriopore.
- Sexes are separate.
- External fertilization.

Amphioxus Maintains All Chordate Characters

- Notochord
- Dorsal neural tube
- Pharyngeal slits/pouches originate from neural crest
- Postanal tail
- Endostyle



Amphioxus characteristics not shared with vertebrates

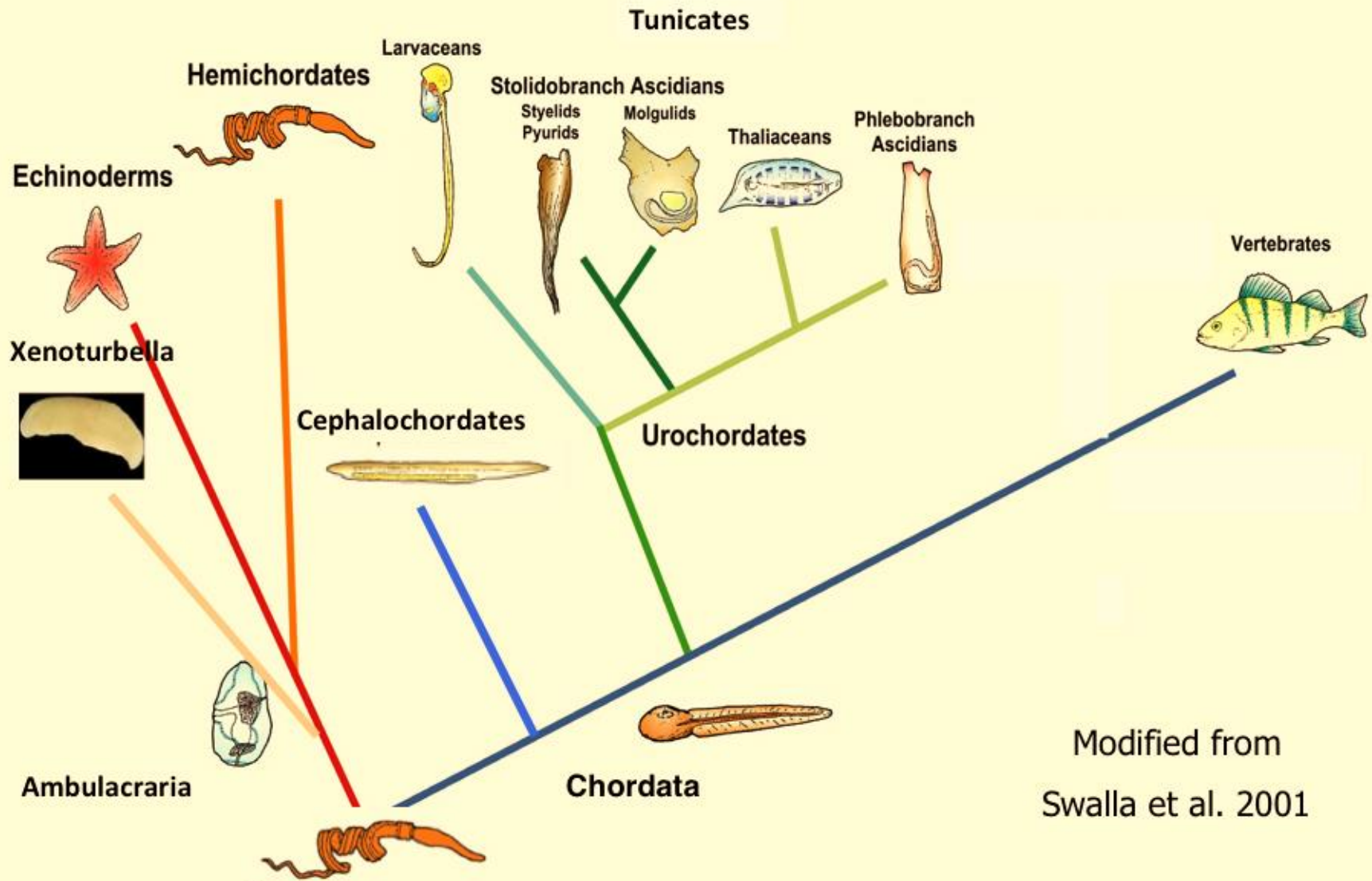
- Amphioxus however lacks several characteristics that biologists think the ancestor of vertebrates possessed. These include:
 - Tripartite brain (with forebrain, midbrain and hindbrain) protected by a cranium (skull)
 - Chambered heart
 - Muscular gut and pharynx (food moved through gut by ciliary action not peristalsis)

Amphioxus characteristics not shared with vertebrates

- Various special sensory organs (eyes, chemical and pressure receptors, nose, inner ear)
- Neural crest (ectodermal cells that are found on the embryonic neural tube and are engaged in the formation of the cranium, tooth dentine, some endocrine glands and Schwann cells, which provide myelin insulation to nerve cells).

Ancestry and Evolution of Chordates

Deuterostome Evolution





The end